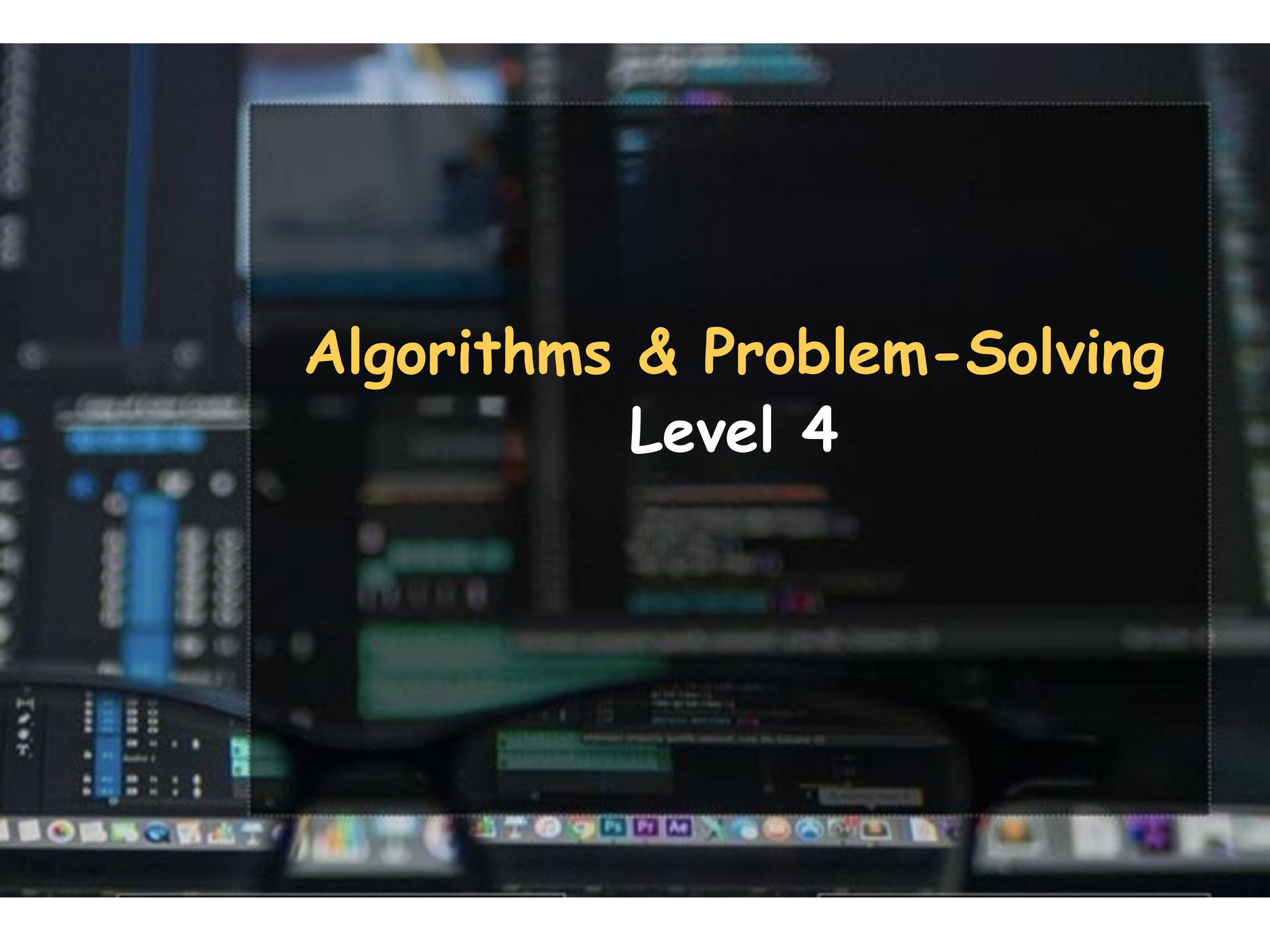
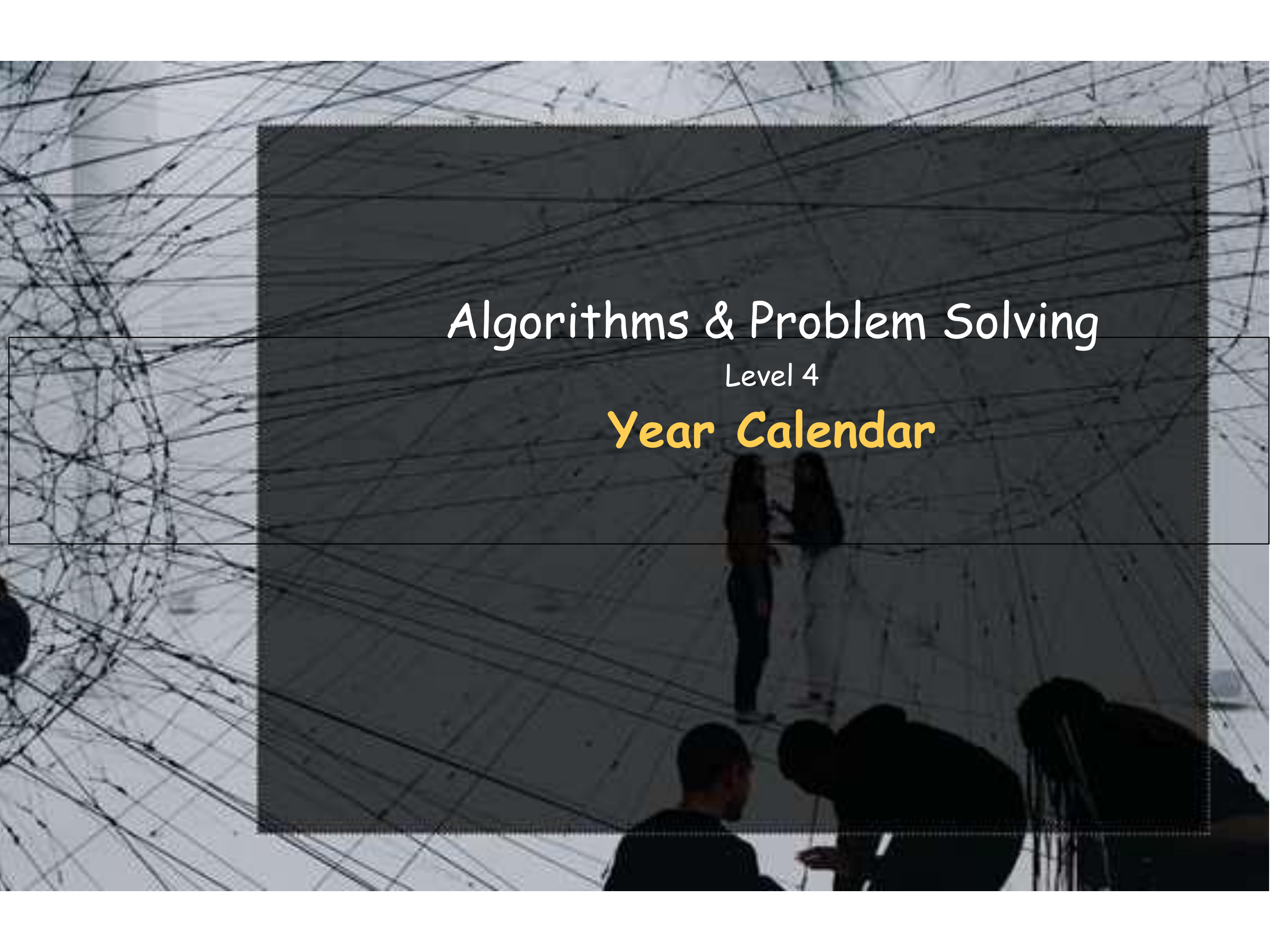




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Year Calendar



Definition

- The program uses the same logic as the monthly calendar (in Problem #08), but here the function for printing each month is called within a loop that goes through all 12 months of the year.

Procedures

① The ReadYear()i function

- asks the user to enter the year

② The IsLeapYear(int year) function

- Determines if a year is a leap year:

- A year is a leap year if:

- $$(year \% 4 == 0 \ \&\& \ year \% 100 != 0) \ || \ (year \% 400 == 0)$$

③ The NumberOfDaysInAMonth(int month, int year) function

- Returns the number of days in each month (adjusting February if it's a leap year)

③ The NumberOfDaysInAMonth(int month, int year) function

- Returns the number of days in each month (adjusting February if it's a leap year)

④ The DayOfWeekOrder(int day, int month, int year) function

- Calculating the day of the week order using Zeller's Congruence:

- $$(day + y + (y/4) - (y/100) + (y/400) + ((31 * m) / 12)) \% 7$$

- The result is 0 to 6 (0 = Sun, 1 = Mon, ..., 6 = Sat)

⑤ The DayShortName(int dayOrder) function

- Returns the abbreviation of the day name, such as:

- "Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat"

⑥ The MonthShortName(int monthNumber) function returns the abbreviated name of the month, such as:

"Jan", "Feb", "Mar", ..., "Dec"

7 The PrintMonthCalendar(int month, int year) function:

- Prints the days of the month in the correct format according to the first day of the month:

- Specify the first day of the week

- Print headings (Sun, Mon, Tue, ...)

- Leave spaces before the start of the month

- Print the days in their order within the weeks.

8 The PrintYearCalendar(int year) function

- Prints all months of the year in order by calling PrintMonthCalendar in a loop:

```
for (int i = 1; i <= 12; i++)
```

```
    PrintMonthCalendar(i, year);
```

Example:

Please enter a year to check? 1981
Calendar - 1981

Jan						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3				
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Feb						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28



Note:

- A leap year only affects the number of days in February (29 instead of 28)
- The week begins on Sunday
- There is no invalid year entry check

Explanation:

- The program uses the `DayOfWeekOrder` function to accurately calculate the beginning of each month using mathematical operations instead of pre-made time libraries.

Summary

- ✓ The program uses:
 - Looping for printing months
 - Modular functions for task division
 - Neat output formatting using printf

Thank you 😊